

PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL (Create Index, Create View, Analyze Table)



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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1. Design database untuk suatu Perusahaan

a. Tabel Departmen

Query :

```
CREATE TABLE departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(50) NOT NULL,  
    manager_id INT,  
    location_id INT  
);
```

Hasil Query :

	↕ COLUMN_NAME	↕ DATA_TYPE	↕ NULLABLE	DATA_DEFAULT	↕ COLUMN_ID	↕
1	DEPARTMENT_ID	NUMBER(38,0)	No	(null)	1	(
2	DEPARTMENT_NAME	VARCHAR2(50 BYTE)	No	(null)	2	(
3	MANAGER_ID	NUMBER(38,0)	Yes	(null)	3	(
4	LOCATION_ID	NUMBER(38,0)	Yes	(null)	4	(

b. Tabel Employees

Query :

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100),  
    salary DECIMAL(10, 2),  
    commission_pct DECIMAL(5, 2),  
    hire_date DATE NOT NULL,  
    department_id INT,  
    CONSTRAINT fk_department  
        FOREIGN KEY (department_id)  
        REFERENCES departments(department_id)  
);
```

Hasil Query :

	⚡ COLUMN_NAME	⚡ DATA_TYPE	⚡ NULLABLE	DATA_DEFAULT	⚡ COLUMN_ID	⚡
1	EMPLOYEE_ID	NUMBER(38,0)	No	(null)	1	(
2	LAST_NAME	VARCHAR2(50 BYTE)	No	(null)	2	(
3	EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	3	(
4	SALARY	NUMBER(10,2)	Yes	(null)	4	(
5	COMMISSION_PCT	NUMBER(5,2)	Yes	(null)	5	(
6	HIRE_DATE	DATE	No	(null)	6	(
7	DEPARTMENT_ID	NUMBER(38,0)	Yes	(null)	7	(

c. Tabel Suppliers

Query :

```
CREATE TABLE suppliers (  
    supplier_id INT PRIMARY KEY,  
    supplier_name VARCHAR(100) NOT NULL,  
    contact_name VARCHAR(50),  
    phone VARCHAR(20)  
);
```

Hasil Query :

SUPPLIER...	SUPPLIER...	CONTACT...	PHONE
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Tabel products

Query :

```
CREATE TABLE products (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(100) NOT NULL,  
    supplier_id INT,  
    unit_price DECIMAL(10, 2),  
    stock_quantity INT,  
    CONSTRAINT fk_supplier  
        FOREIGN KEY (supplier_id)  
        REFERENCES suppliers(supplier_id)  
);
```

Hasil Query :

PRODUCT...	PRODUCT...	SUPPLIER...	UNIT_PRICE	STOCK_Q...
------------	------------	-------------	------------	------------

2. Analisis Table Menggunakan Compute Statistic

Query :

```
--2  
ANALYZE TABLE departments COMPUTE STATISTICS;  
ANALYZE TABLE employees COMPUTE STATISTICS;
```

Hasil Query :

Table DEPARTMENTS analyzed.

Table EMPLOYEES analyzed.

3. Isi data untuk table

Table Departments

Query :

```
--3
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (10, 'Administration', 200, 1700);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (20, 'Marketing', 201, 1800);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (30, 'Purchasing', 202, 1700);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (40, 'Human Resources', 203, 2400);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (50, 'Shipping', 204, 1500);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (60, 'IT', 205, 1400);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (70, 'Public Relations', 206, 2700);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (80, 'Sales', 207, 2500);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (90, 'Executive', 208, 1700);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (100, 'Finance', 209, 1700);
```

Hasil Query :

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	202	1700
4	40	Human Resources	203	2400
5	50	Shipping	204	1500
6	60	IT	205	1400
7	70	Public Relations	206	2700
8	80	Sales	207	2500
9	90	Executive	208	1700
10	100	Finance	209	1700

Tabel Employees

Query :

```
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (1, 'King', 'king@mail.com', 5000, 0.1, TO_DATE('2023-01-01', 'YYYY-MM-DD'), 90);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (2, 'Kochhar', 'kochhar@mail.com', 4500, NULL, TO_DATE('2023-02-15', 'YYYY-MM-DD'), 90);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (3, 'Abel', 'abel@mail.com', 3800, 0.2, TO_DATE('2023-03-10', 'YYYY-MM-DD'), 80);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (4, 'Taylor', 'taylor@mail.com', 3200, 0.15, TO_DATE('2023-04-05', 'YYYY-MM-DD'), 80);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (5, 'Grant', 'grant@mail.com', 2800, 0.1, TO_DATE('2023-05-20', 'YYYY-MM-DD'), 80);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (6, 'Hunold', 'hunold@mail.com', 4000, NULL, TO_DATE('2023-01-20', 'YYYY-MM-DD'), 60);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (7, 'Ernst', 'ernst@mail.com', 3500, NULL, TO_DATE('2023-02-10', 'YYYY-MM-DD'), 60);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (8, 'Lorentz', 'lorentz@mail.com', 3000, NULL, TO_DATE('2023-03-01', 'YYYY-MM-DD'), 60);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (9, 'Whalen', 'whalen@mail.com', 2500, NULL, TO_DATE('2023-04-12', 'YYYY-MM-DD'), 10);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (10, 'Zlotkey', 'zlotkey@mail.com', 3100, 0.1, TO_DATE('2023-05-01', 'YYYY-MM-DD'), 80);
```

Hasil Query :

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPART
1	1	King	king@mail.com	5000	0,1	01-01-2023	
2	2	Kochhar	kochhar@mail.com	4500	(null)	15-02-2023	
3	3	Abel	abel@mail.com	3800	0,2	10-03-2023	
4	4	Taylor	taylor@mail.com	3200	0,15	05-04-2023	
5	5	Grant	grant@mail.com	2800	0,1	20-05-2023	
6	6	Hunold	hunold@mail.com	4000	(null)	20-01-2023	
7	7	Ernst	ernst@mail.com	3500	(null)	10-02-2023	
8	8	Lorentz	lorentz@mail.com	3000	(null)	01-03-2023	
9	9	Whalen	whalen@mail.com	2500	(null)	12-04-2023	
10	10	Zlotkey	zlotkey@mail.com	3100	0,1	01-05-2023	

4. Analisis Table data setelah di isi data

Query :
<pre>ANALYZE TABLE departments COMPUTE STATISTICS; ANALYZE TABLE employees COMPUTE STATISTICS;</pre>
Hasil Query :
<pre>Table DEPARTMENTS analyzed. Table EMPLOYEES analyzed.</pre>

5. View Empvu80

Query :
<pre>CREATE OR REPLACE VIEW empvu80 AS SELECT employee_id AS id_number, last_name AS name, salary, department_id FROM employees WHERE department_id = 80;</pre>
Hasil Query :
<pre>View EMPVU80 created.</pre>

6. Tampilkan empvu80

Query :																		
<pre>DESC empvu80;</pre>																		
Hasil Query :																		
<table><tr><th>Name</th><th>Null?</th><th>Type</th></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>ID_NUMBER</td><td>NOT NULL</td><td>NUMBER(38)</td></tr><tr><td>NAME</td><td>NOT NULL</td><td>VARCHAR2(50)</td></tr><tr><td>SALARY</td><td></td><td>NUMBER(10,2)</td></tr><tr><td>DEPARTMENT_ID</td><td></td><td>NUMBER(38)</td></tr></table>	Name	Null?	Type	-----	-----	-----	ID_NUMBER	NOT NULL	NUMBER(38)	NAME	NOT NULL	VARCHAR2(50)	SALARY		NUMBER(10,2)	DEPARTMENT_ID		NUMBER(38)
Name	Null?	Type																
-----	-----	-----																
ID_NUMBER	NOT NULL	NUMBER(38)																
NAME	NOT NULL	VARCHAR2(50)																
SALARY		NUMBER(10,2)																
DEPARTMENT_ID		NUMBER(38)																

7. Non unique-index pada kolom FK pada table employee

Query :
<pre>CREATE INDEX emp_dept_idx ON employees(department_id);</pre>
Hasil Query :
<pre>Index EMP_DEPT_IDX created.</pre>

Link video : [Link Record](#)